1 Two identical springs, each with a spring constant k, are attached to each other. The spring combination is attached to a wooden block and a string. The string is also attached to the spindle of a d.c. motor, as shown in Figure 1.1.

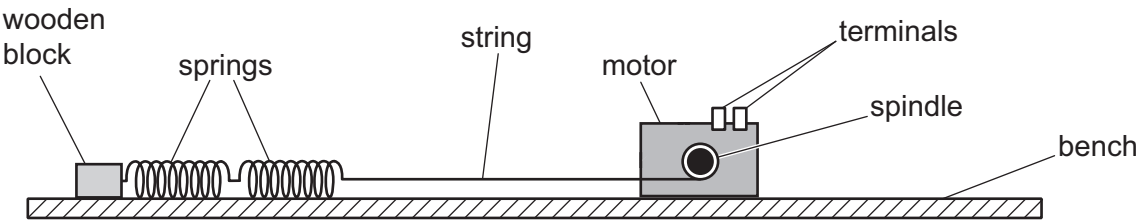

Figure 1.1

The terminals of the motor are connected to a capacitor of capacitance C.

The capacitor is initially charged by applying a potential difference V across it. The capacitor is then discharged through the motor.

As the spindle of the motor turns, the string wraps around the spindle causing the spring combination to extend. The maximum extension of the spring combination is e.

It is suggested that e is related to C by the relationship

$$\frac{ke^2}{2} = \beta CV^2$$

where β is a constant.

Plan a laboratory experiment to test the relationship between e and C.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine a value for β.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

DO NOT WRITE IN THIS MARGIN